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| Total Members | 1 |
| Members | |

| Name | Role | Email |
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## Overview

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| Project | ShockBook - Quantitative Crypto Portfolio Stress-Test Engine & MCP Server |
| Track | Track 1 (AI Agent), Track 2 (Operator Cockpit), Track 3 (MCP Skill Server) |
| Problem | Single-token tools evaluate assets in isolation or two at a time. On a multi-token book (SOL, AVAX, BTC, USDC), bullish single-token scores hide 168-hour cross-asset correlation spikes (r = 0.89), exposing traders to severe unhedged tail drawdowns. |
| Solution | ShockBook fuses RYO-CHAN token and regime intelligence with 168-hour OKX 1H candles and River AI (DeepSeek-V4.1-Flash-1017) to project 1-sigma/2-sigma shock cones, compute correlation-capped USDC rebalance trims, and commit SHA-256 verifiable mandates. |
| Key Features | 1) /desk: 24h 1s/2s Shock Cone, Sankey rebalance solver, 168h correlation matrix & waterfall. 2) /ledger: SQLite WAL mandate ledger, SHA-256 replay & counterfactual P&L. 3) /skills: MCP JSON-RPC 2024-11-05 server with 60 RPM guard & null safety. |
| Target Users | Multi-token crypto traders, DeFi treasury managers, quantitative risk analysts, and autonomous AI agents consuming RYO-CHAN Model Context Protocol (MCP) tools. |
| Scope | Multi-token portfolio covariance & shock simulation (/desk), immutable SQLite WAL mandate audit ledger (/ledger), and native MCP JSON-RPC 2024-11-05 risk skills (portfolio_stress_test and correlation_guard) with resilience fault drills (/skills). |
| Limitations | Advisory simulation & rebalance mandate engine (does not custody private keys or auto-broadcast on-chain swaps). RYO-CHAN MCP enforces a 60 RPM limit, handled via ShockBook sliding-window quota guard and SQLite WAL snapshot fallback. |

## Tech Stack

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| Frontend Stack | Semantic HTML5, Architectural Chiaroscuro CSS Grid (Dual Obsidian/Alabaster Polarity), Vanilla ES2022 JS, Interactive SVG Charts (Shock Cone, Sankey, Covariance Heatmap), HTML5 2D Topo Canvas Shader, Self-Hosted WOFF2 Fonts. |
| Backend Stack | Python 3.11+, FastAPI, Starlette ASGI, SQLite 3 (WAL mode, synchronous=NORMAL, foreign_keys=ON), Pydantic v2, HTTPX Client, Canonical JSON + SHA-256 Deterministic Replay Engine, MCP JSON-RPC 2024-11-05 Server. |
| AI Model(s) | River AI (deepseek-ai/DeepSeek-V4.1-Flash-1017 via https://api.river.ai/v1) for Risk Officer cause-and-effect audit synthesis, paired with a deterministic quantitative rule fallback when offline or rate-limited. |
| Other Technologies | RYO-CHAN MCP Server (https://app-ryochan.com/api/mcp: analyze_token, deep_analysis, market_overview, monitor_market_sentiment_shift), OKX Public V5 API (168h 1H OHLCV candles), Vercel Serverless Python, Pytest. |

## Repository / Demo

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| Github Repository | https://github.com/RYO-Digital/ryochan-hackathon_repository-239 (Public Mirror: https://github.com/Pizzylee001/ShockBook) |
| Demo Video | https://youtu.be/X8mrggIA28M |
| Documentation | Live App: https://shockbook.vercel.app \| Full README.md, DESIGN.md, and docs/diagrams/architecture.html in repository. |

## Testing Information

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| How to run the Project | Evaluate directly at https://shockbook.vercel.app (/desk, /ledger, /skills) or run locally via uvicorn main:app --host 0.0.0.0 --port 8000. |
| Prerequisites | Python 3.11+, pip, RYO-CHAN MCP Credential (RYO_MCP_KEY), and River AI API Key (RIVER_API_KEY, optional due to deterministic fallback). |
| Installation Steps | 1) git clone https://github.com/RYO-Digital/ryochan-hackathon_repository-239.git 2) cd ryochan-hackathon_repository-239 3) pip install -r requirements.txt 4) cp .env.example .env |
| Environment Variables | RYO_MCP_URL=https://app-ryochan.com/api/mcp \| RYO_MCP_KEY=<credential> \| RIVER_BASE_URL=https://api.river.ai/v1 \| RIVER_MODEL=deepseek-ai/DeepSeek-V4.1-Flash-1017 \| RIVER_API_KEY=<key> |
| Build Command | pip install -r requirements.txt (Zero frontend build step required; self-hosted static WOFF2 fonts and SVG/Canvas assets) |
| Run Command | uvicorn main:app --host 0.0.0.0 --port 8000 |

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| Test Command | pytest tests/test_shockbook.py -v (8 automated tests verifying 168h covariance math, explicit null preservation, SQLite WAL SHA-256 replay, MCP JSON-RPC 2024-11-05 & REST fault drills, routes, and secret hygiene) |
| Test Account(s) | No login required. Open access at https://shockbook.vercel.app (/desk, /ledger, /skills) and MCP JSON-RPC endpoint at POST https://shockbook.vercel.app/api/mcp/rpc (Hackathon Key ID 9aa2de95 configured on server). |